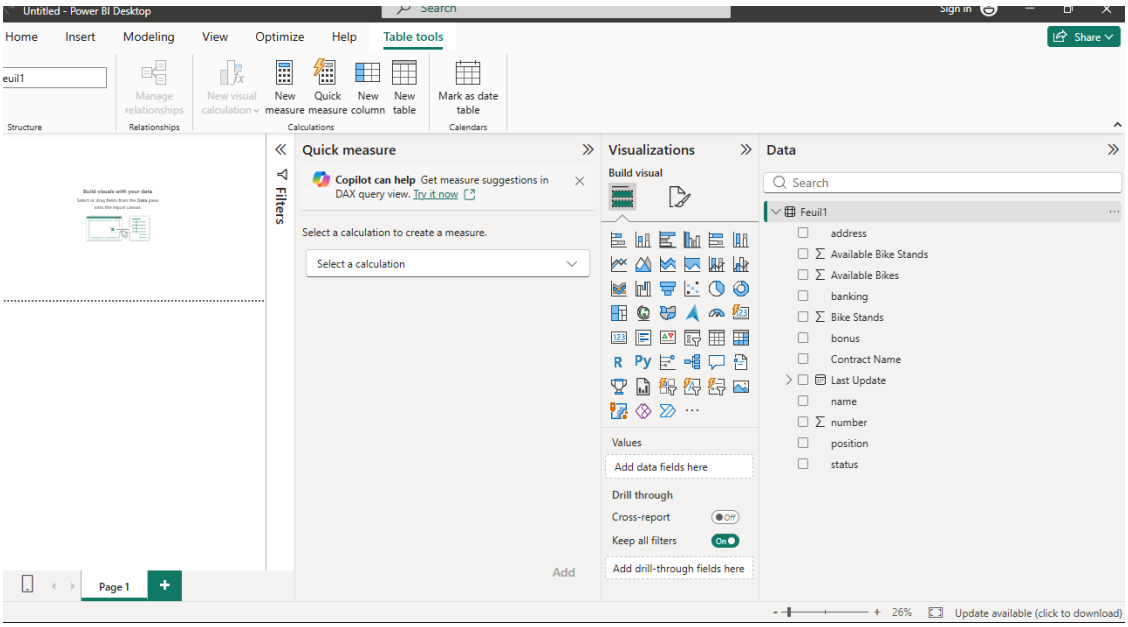
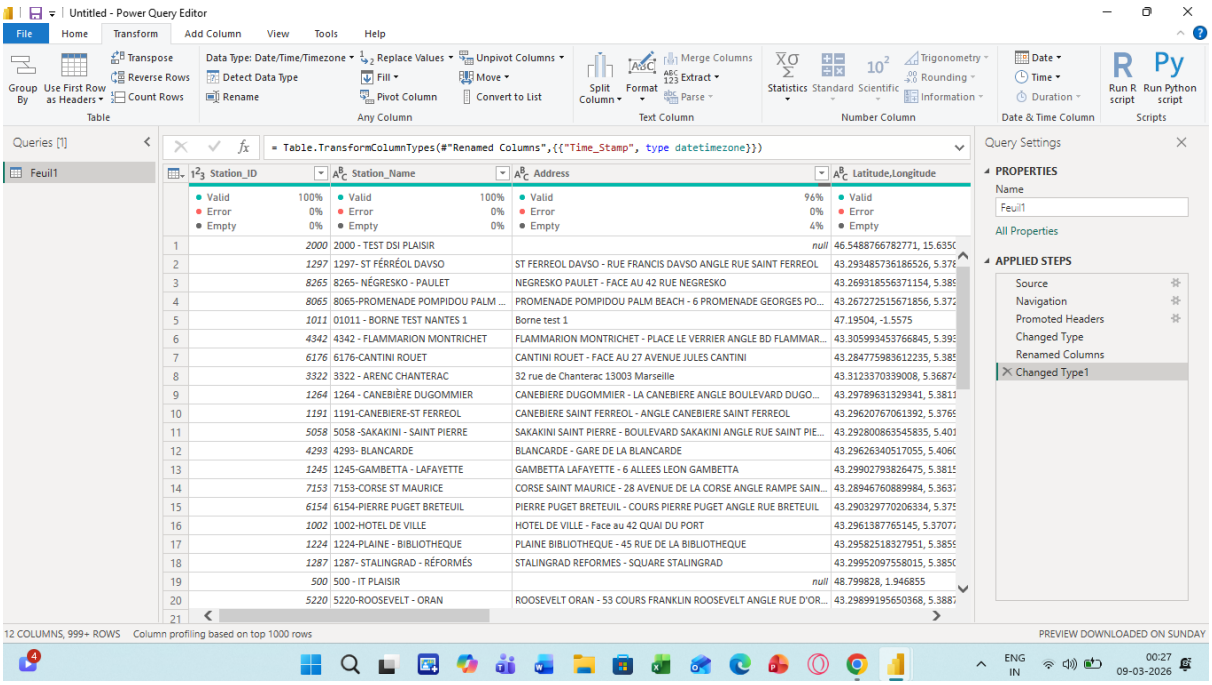
Analyzing Smart City Bike Sharing Data using Power BI

1)Import and explore the dataset in Power BI.

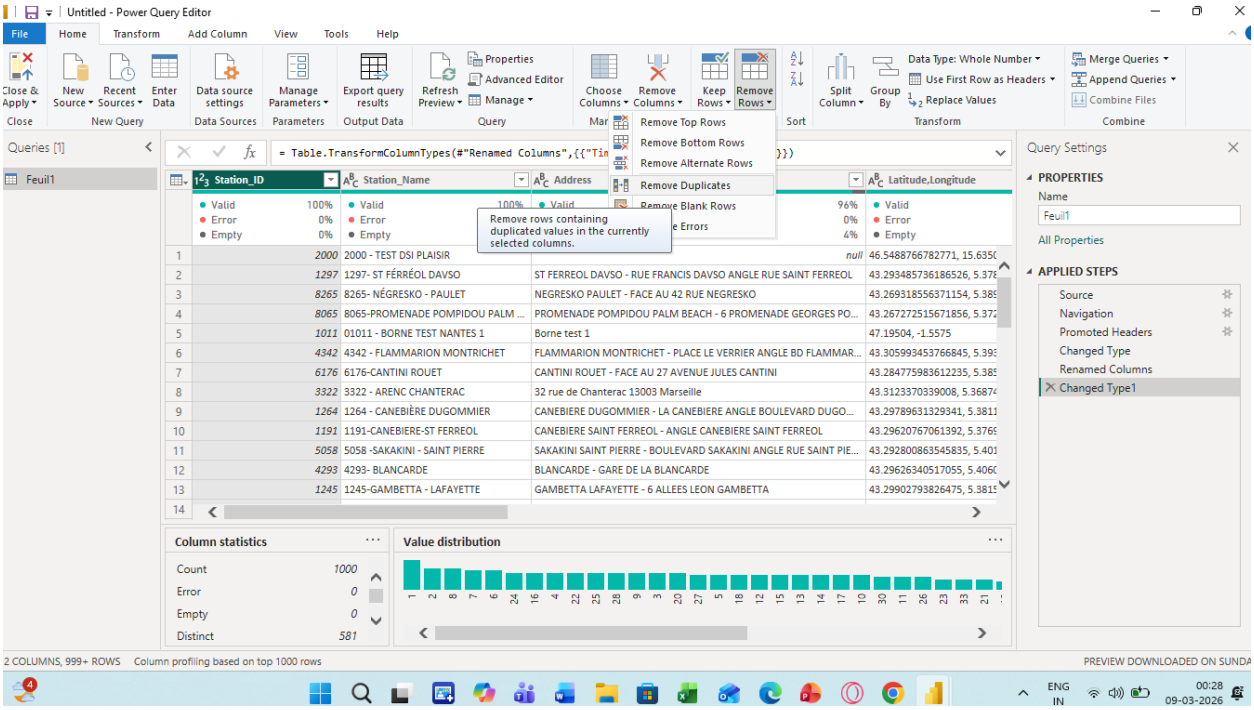


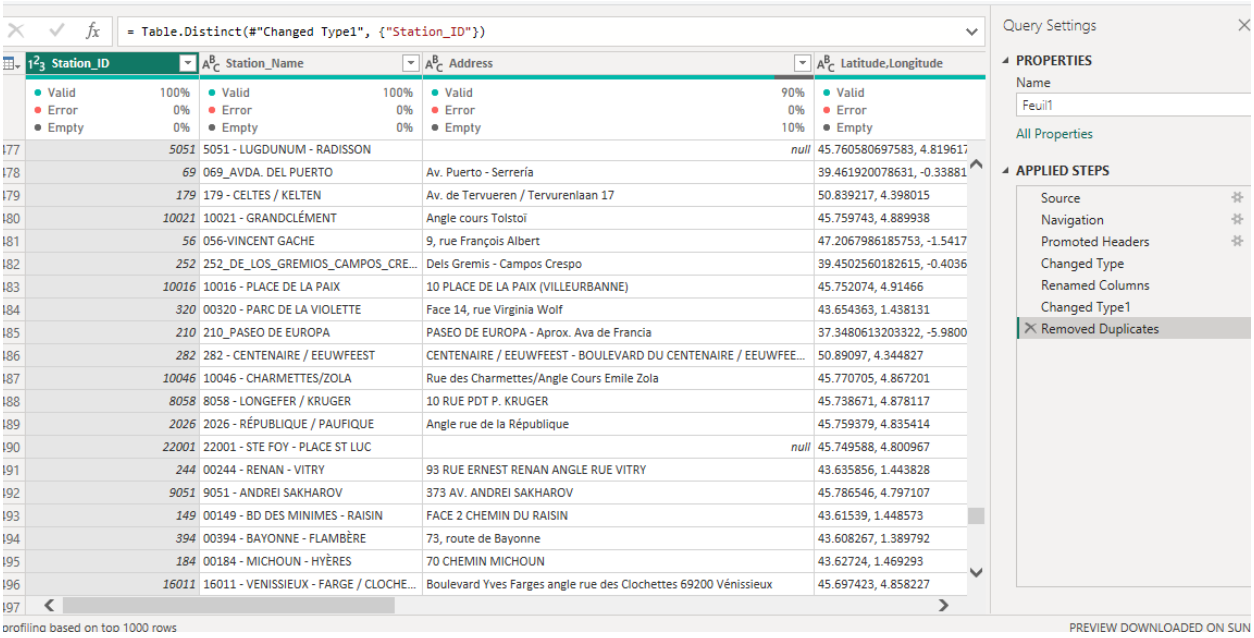
2)Clean and transform data using Power Query

Transformed data into Power Query reanmed the header and changed data type in Power Query.

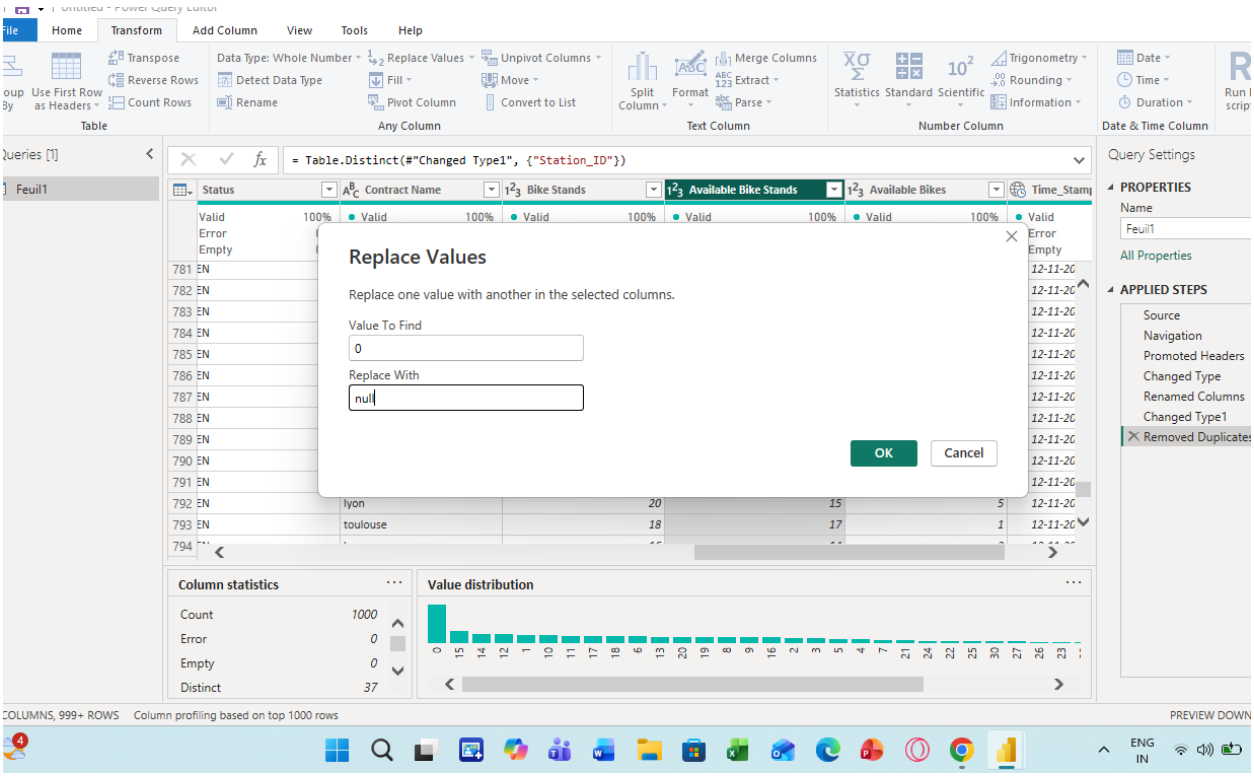


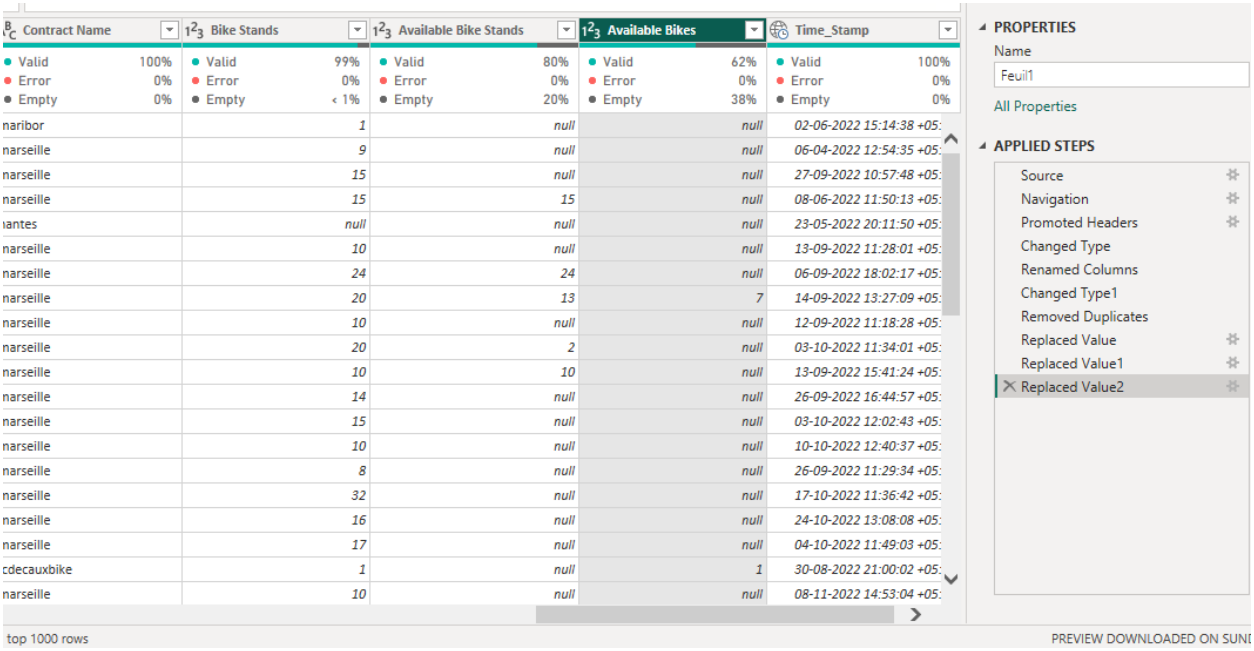
Removed Duplicates if any



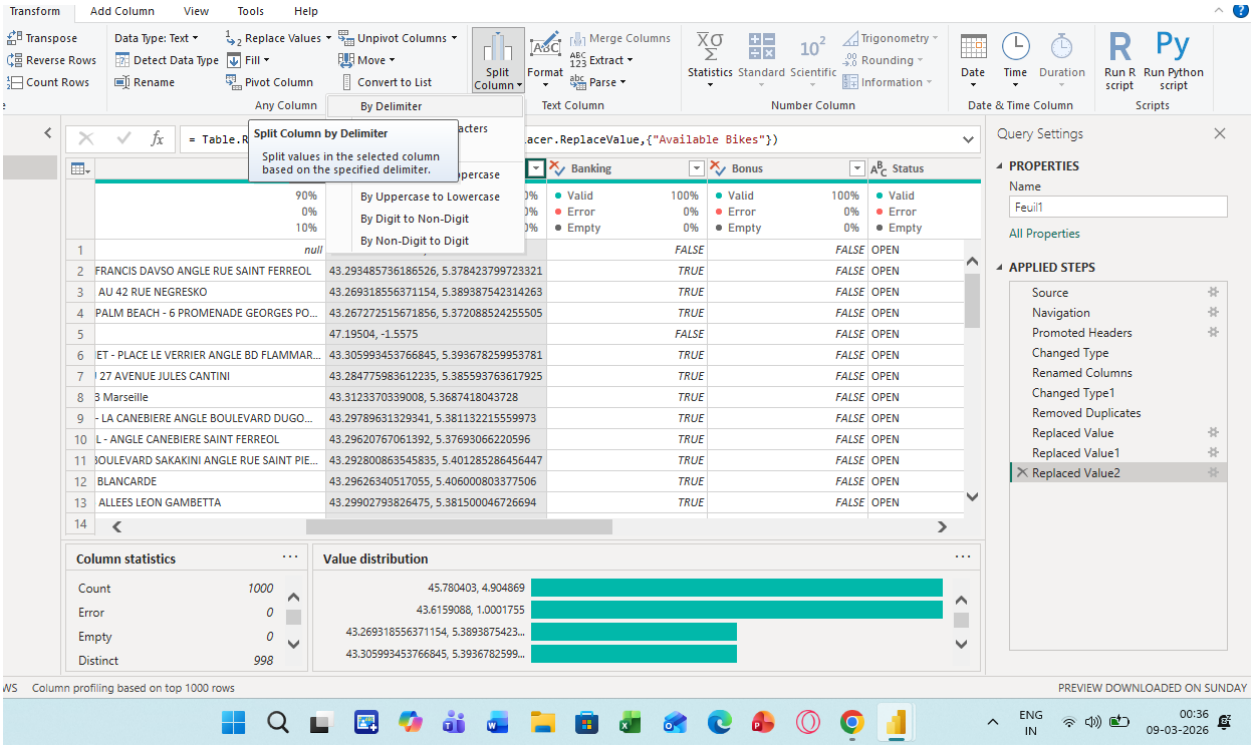


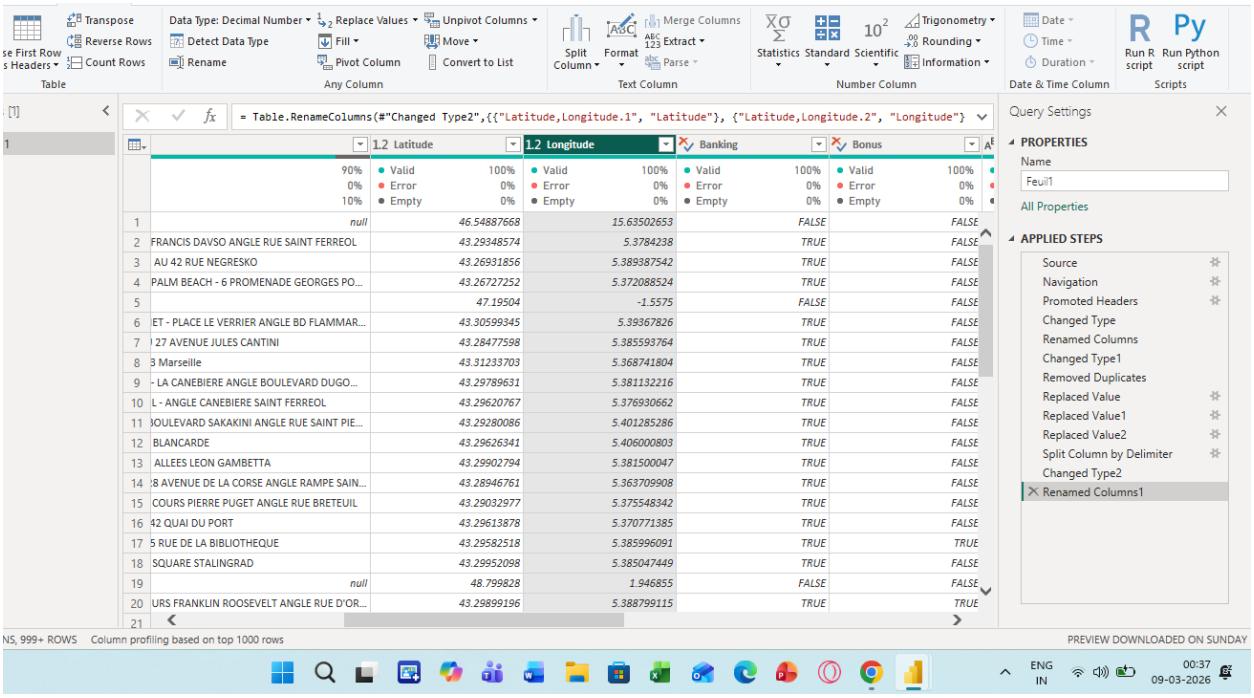
Replace null with 0



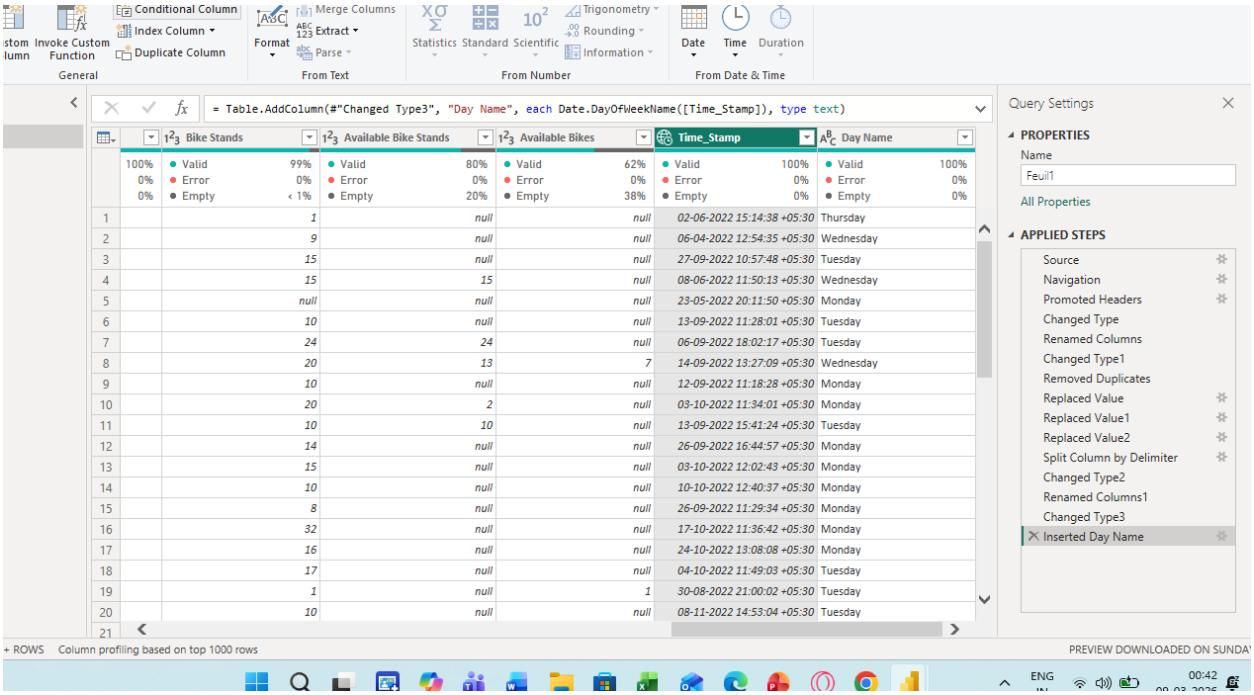


Seperated Latitude and Longitude



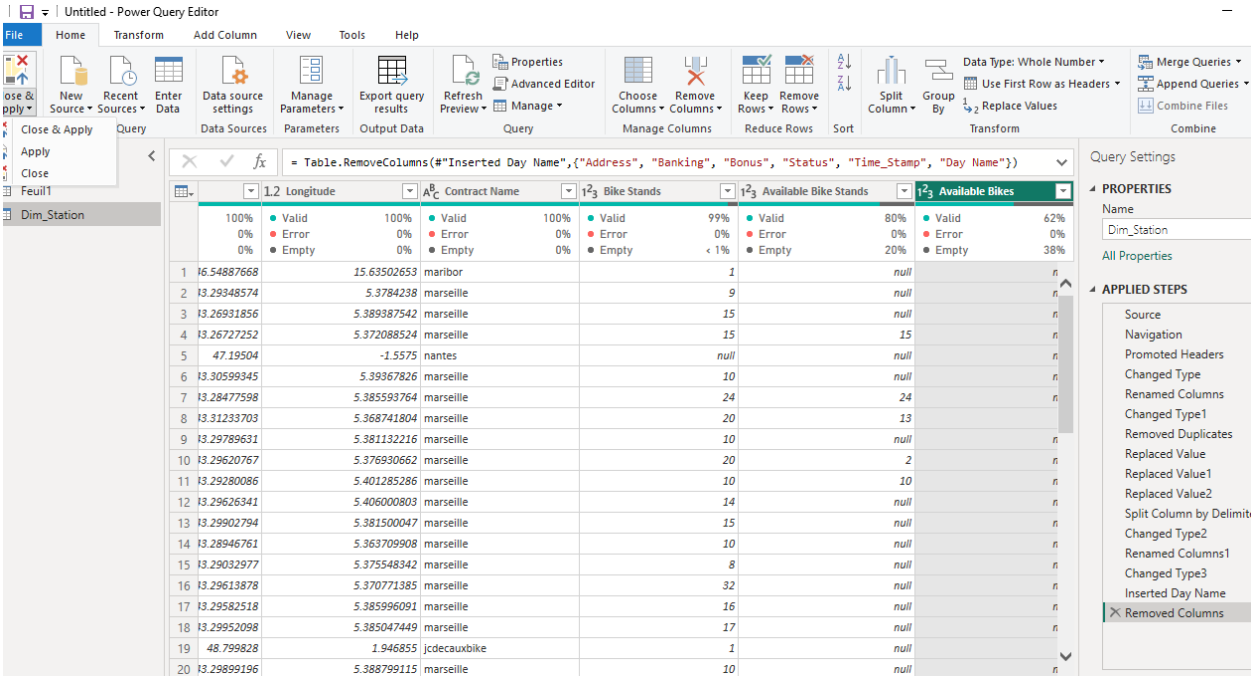


Created calculated column

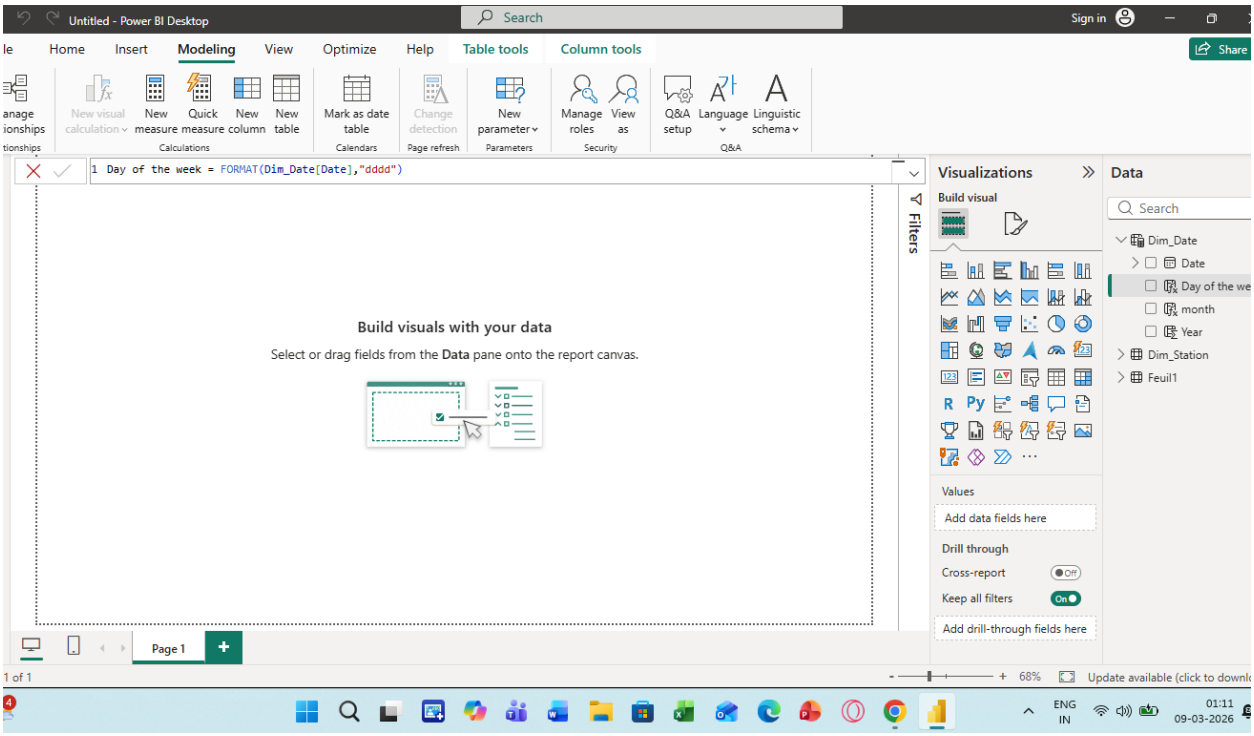


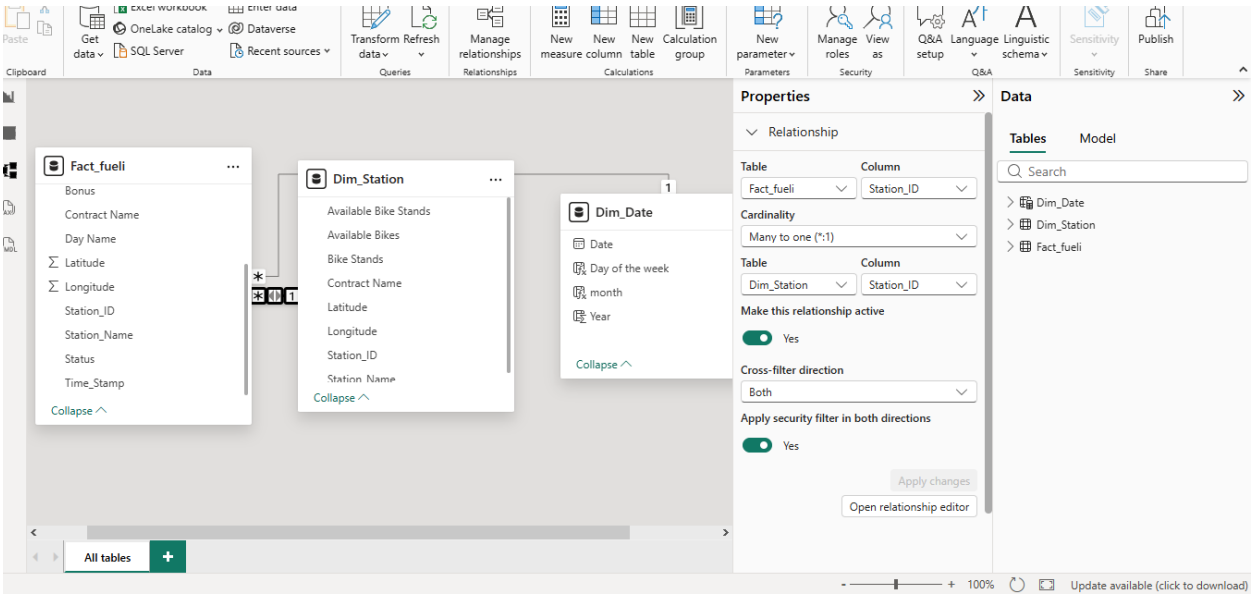
Logical data model.

Created Dimensional table Dim_Station in Power Query



Created Dimensional table Dim_Table by Dax measure in Power BI


Untitled - Power BI Desktop
Search
Sign in
Home
Insert
Modeling
View
Optimize
Help
Table tools
Column tools
Share
New measure
Quick measure
New column
New table
Mark as date table
Change detection
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Calculations
Calendars
Page refresh
Parameters
Security
Q&A
1 Day of the week = FORMAT(Dim_Date[Date],"dddd")
Build visuals with your data
Select or drag fields from the Data pane onto the report canvas.
Page 1
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Filters
Data
Search
Dim_Date
Date
Day of the we
month
Year
Dim_Station
Feuil1
1 of 1
68%
Update available (click to downl


Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
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Q&A setup
Language
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Sensitivity
Publish
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
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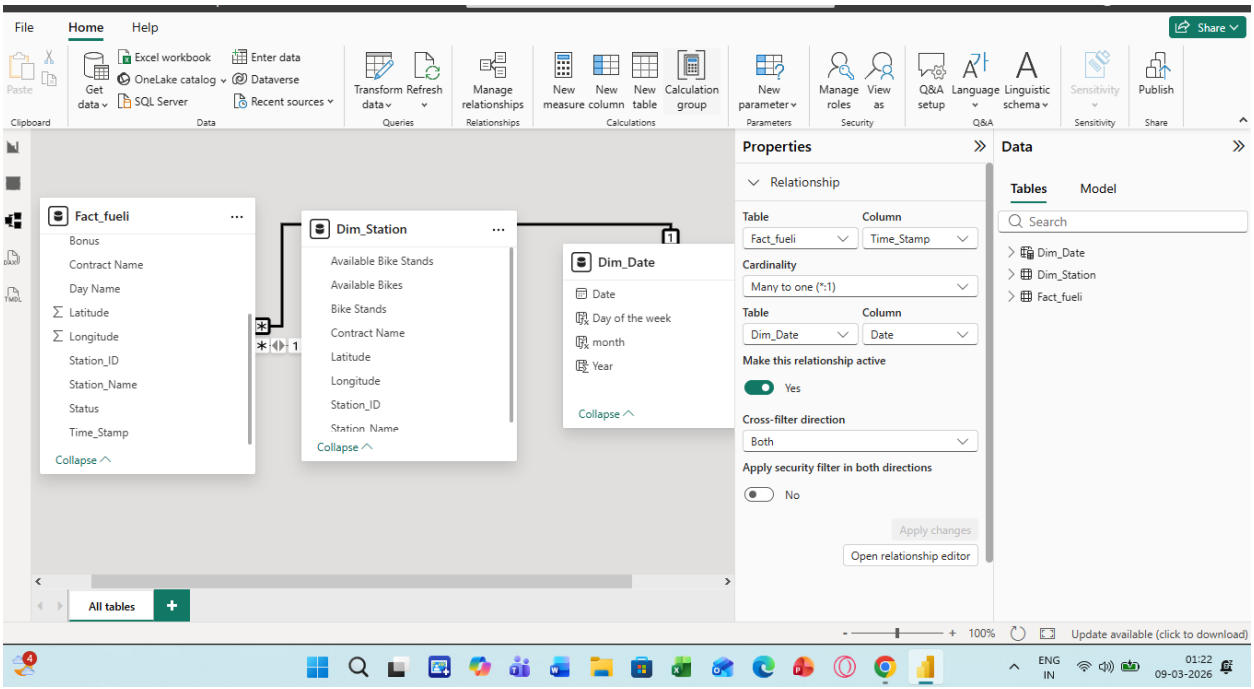
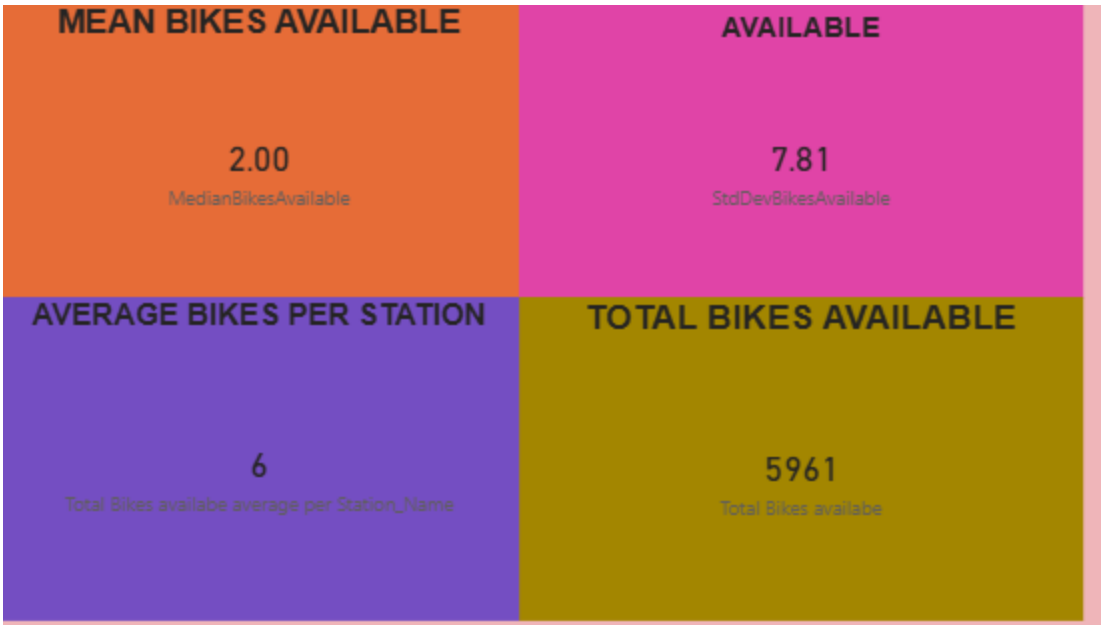
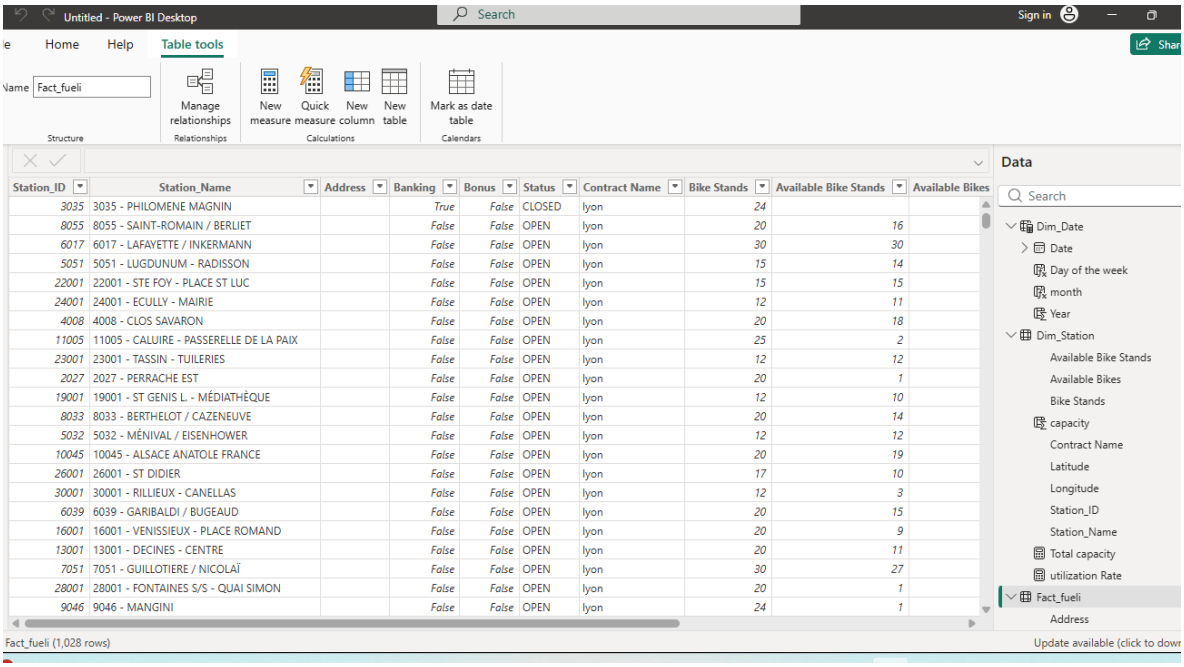
Share
Fact_fueli
Bonus
Contract Name
Day Name
Latitude
Longitude
Station_ID
Station_Name
Status
Time_Stamp
Collapse
Dim_Station
Available Bike Stands
Available Bikes
Bike Stands
Contract Name
Latitude
Longitude
Station_ID
Station_Name
Collapse
Dim_Date
Date
Day of the week
month
Year
Collapse
Properties
Relationship
Table
Column
Fact_fueli
Station_ID
Cardinality
Many to one (*:1)
Table
Column
Dim_Station
Station_ID
Make this relationship active
Yes
Cross-filter direction
Both
Apply security filter in both directions
Yes
Apply changes
Open relationship editor
Data
Tables
Model
Search
Dim_Date
Dim_Station
Fact_fueli
All tables
100%
Update available (click to download)



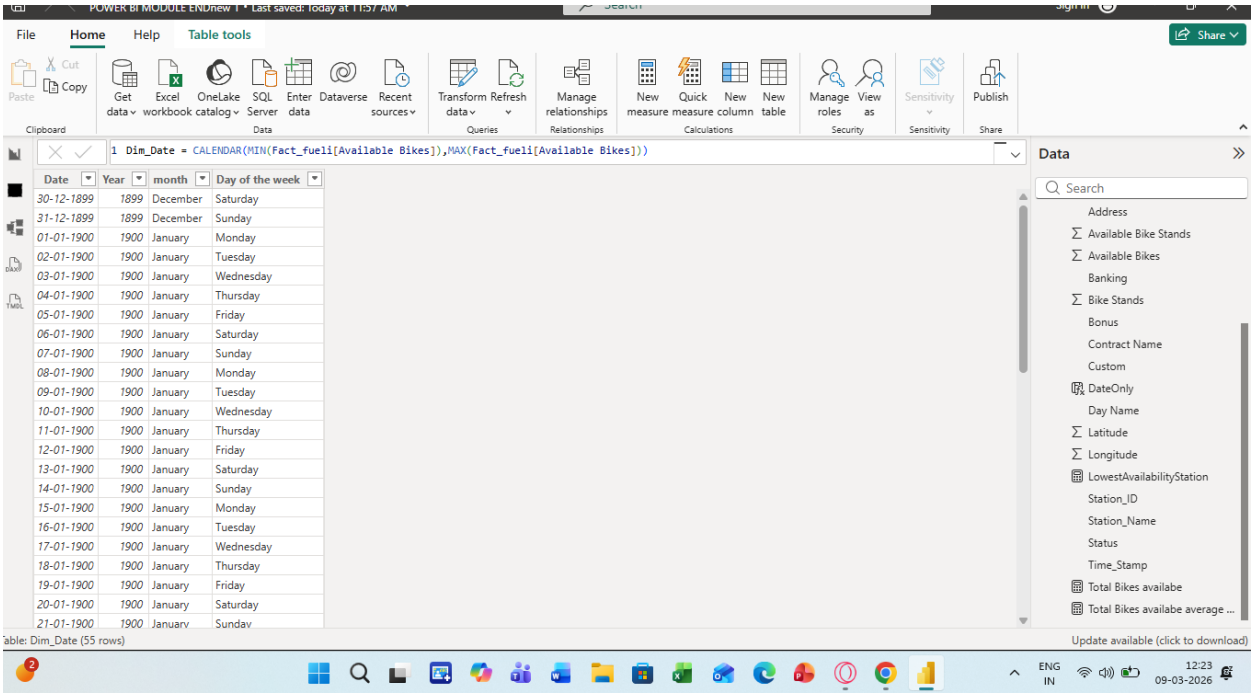
Creating DAX Measure

```
Total capacity = SUM(Dim_Station[capacity])
utilization Rate = DIVIDE(SUM(Fact_fueli[Available Bikes]),SUM(Dim_Station[capacity]))
Total Bikes availabe = SUM(Fact_fueli[Available Bikes])
Total Bikes availabe average per Station_Name =
AVERAGEX(KEEPFILTERS(VALUES('Dim_Station'[Station_Name])),CALCULATE([Total Bikes availabe]))
LowestAvailabilityStation = VAR StationTable =TOPN(1, SUMMARIZE(Fact_fueli,
Fact_fueli[Station_Name], "Avail", SUM(Fact_fueli[Available Bikes])), [Avail], ASC)RETURN
MAXX(Fact_fueli, Fact_fueli[Station_Name])
MedianBikesAvailable = MEDIAN(Fact_fueli[Available Bikes])
StdDevBikesAvailable = STDEV.P(Fact_fueli[Available Bikes])
```





Station_ID	Station_Name	Address	Banking	Bonus	Status	Contract Name	Bike Stands	Available Bike Stands	Available Bikes
3035	3035 - PHILOMENE MAGNIN		True	False	CLOSED	lyon	24		
8055	8055 - SAINT-ROMAIN / BERLIET		False	False	OPEN	lyon	20	16	
6017	6017 - LAFAYETTE / INKERMANN		False	False	OPEN	lyon	30	30	
5051	5051 - LUGDUNUM - RADISSON		False	False	OPEN	lyon	15	14	
22001	22001 - STE FOY - PLACE ST LUC		False	False	OPEN	lyon	15	15	
24001	24001 - ECULLY - MAIRIE		False	False	OPEN	lyon	12	11	
4008	4008 - CLOS SAVARON		False	False	OPEN	lyon	20	18	
11005	11005 - CALUIRE - PASSERELLE DE LA PAIX		False	False	OPEN	lyon	25	2	
23001	23001 - TASSIN - TUILERIES		False	False	OPEN	lyon	12	12	
2027	2027 - PERRACHE EST		False	False	OPEN	lyon	20	1	
19001	19001 - ST GENIS L. - MÉDIATHÈQUE		False	False	OPEN	lyon	12	10	
8033	8033 - BERTHELOT / CAZENEUVE		False	False	OPEN	lyon	20	14	
5032	5032 - MÉNIVAL / EISENHOWER		False	False	OPEN	lyon	12	12	
10045	10045 - ALSACE ANATOLE FRANCE		False	False	OPEN	lyon	20	19	
26001	26001 - ST DIDIER		False	False	OPEN	lyon	17	10	
30001	30001 - RILLIEUX - CANELLAS		False	False	OPEN	lyon	12	3	
6039	6039 - GARIBALDI / BUGEAUD		False	False	OPEN	lyon	20	15	
16001	16001 - VENISSIEUX - PLACE ROMAND		False	False	OPEN	lyon	20	9	
13001	13001 - DECINES - CENTRE		False	False	OPEN	lyon	20	11	
7051	7051 - GUILLOTIERE / NICOLAÏ		False	False	OPEN	lyon	30	27	
28001	28001 - FONTAINES S/S - QUAI SIMON		False	False	OPEN	lyon	20	1	
9046	9046 - MANGINI		False	False	OPEN	lyon	24	1	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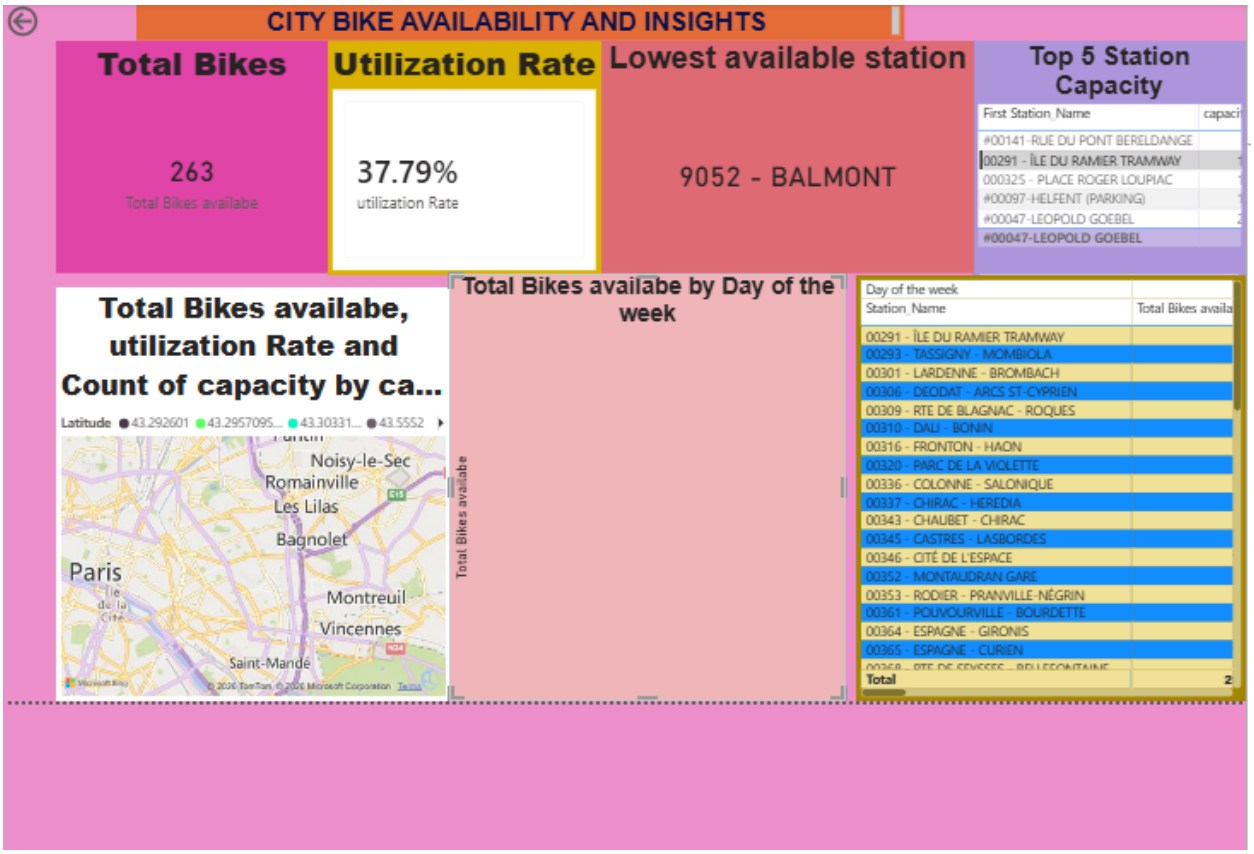
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
Manage relationships
Relationships
New measure
Quick measure
New column
New table
Calculations
Manage roles
View as
Security
Sensitivity
Publish
Share
1 Dim_Date = CALENDAR(MIN(Fact_fueli[Available Bikes]),MAX(Fact_fueli[Available Bikes]))
Date | Year | month | Day of the week
30-12-1899 | 1899 | December | Saturday
31-12-1899 | 1899 | December | Sunday
01-01-1900 | 1900 | January | Monday
02-01-1900 | 1900 | January | Tuesday
03-01-1900 | 1900 | January | Wednesday
04-01-1900 | 1900 | January | Thursday
05-01-1900 | 1900 | January | Friday
06-01-1900 | 1900 | January | Saturday
07-01-1900 | 1900 | January | Sunday
08-01-1900 | 1900 | January | Monday
09-01-1900 | 1900 | January | Tuesday
10-01-1900 | 1900 | January | Wednesday
11-01-1900 | 1900 | January | Thursday
12-01-1900 | 1900 | January | Friday
13-01-1900 | 1900 | January | Saturday
14-01-1900 | 1900 | January | Sunday
15-01-1900 | 1900 | January | Monday
16-01-1900 | 1900 | January | Tuesday
17-01-1900 | 1900 | January | Wednesday
18-01-1900 | 1900 | January | Thursday
19-01-1900 | 1900 | January | Friday
20-01-1900 | 1900 | January | Saturday
21-01-1900 | 1900 | January | Sunday
Data
Search
Address
Available Bike Stands
Available Bikes
Banking
Bike Stands
Bonus
Contract Name
Custom
DateOnly
Day Name
Latitude
Longitude
LowestAvailabilityStation
Station_ID
Station_Name
Status
Time_Stamp
Total Bikes availabe
Total Bikes availabe average ...
Table: Dim_Date (55 rows)
Update available (click to download)
ENG IN
12:23
09-03-2026

DASHBOARD


CITY BIKE AVAILABILITY AND INSIGHTS
Total Bikes
263
Total Bikes availabe
Utilization Rate
37.79%
utilization Rate
Lowest available station
9052 - BALMONT
Top 5 Station Capacity
First Station_Name
capaci
#00141-RUE DU PONT BERELDANGE
00291 - ÎLE DU RAMIER TRAMWAY
000325 - PLACE ROGER LOUPIAC
#00097-HELFENT (PARKING)
#00047-LEOPOLD GOEBEL
#00047-LEOPOLD GOEBEL
Total Bikes availabe, utilization Rate and Count of capacity by ca...
Latitude 43.292601 43.2957095... 43.30331... 43.5552
Noisy-le-Sec
Romainville
Les Lilas
Bagnolet
Paris
Montreuil
Vincennes
Saint-Mandé
Total Bikes availabe by Day of the week
Total Bikes availabe
Day of the week
Station_Name
Total Bikes availa
00291 - ÎLE DU RAMIER TRAMWAY
00293 - TASSIGNY - MOMBIOLA
00301 - LARDENNE - BROMBACH
00306 - DEODAT - ARCS ST-CYPRIEN
00309 - RTE DE BLAGNAC - ROQUES
00310 - DALI - BONIN
00316 - FRONTON - HAON
00320 - PARC DE LA VIOLETTE
00336 - COLONNE - SALONIQUE
00337 - CHIRAC - HEREDIA
00343 - CHAUBET - CHIRAC
00345 - CASTRES - LASBORDES
00346 - CITÉ DE L'ESPACE
00352 - MONTAUDRAN GARE
00353 - RODIER - PRANVILLE-NÉGRIN
00361 - POUVOURVILLE - BOURDETTE
00364 - ESPAGNE - GIRONIS
00365 - ESPAGNE - CURIEN
Total